RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Smart Sort: AI Powered Rapid Grading System for Second-Life Li-ion Cells

Design Thinking Lab Report

Submitted by,

AMRUT R	**1RV24EC025**
ANURAG	**1RV24EC031**
BHAVESH S	**1RV24EC044**
CHANDANA B C	**1RV24EC053**

Under the guidance of

Ms R SINDHU RAJENDRAN

Assistant Professor

Dept. of ECE

RV College of Engineering

In partial fulfillment of the requirements for the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

2025-26

RV COLLEGE OF ENGINEERING®, BENGALURU-59

(*Autonomous institution affiliated to VTU, Belagavi*)

Department of Electronics and Communication Engineering

CERTIFICATE

Certified that the Design Thinking Lab project titled "**Smart Sort: AI Powered Rapid Grading System for Second-Life Li-ion Cells**" is carried out by **Amrut R(1RV24EC025), Anurag (1RV24EC031), Bhavesh S(1RV24EC044), Chandana B C(1RV24EC053)** who are the Bonafide student of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the Design Thinking Lab report deposited in the departmental library. The Design Thinking Lab report has been approved as it satisfies the academic requirements in respect of Design Thinking Lab work prescribed by the institution for the said degree.

Signature of the Mentor	Signature of Head of the Department
Ms. R Sindhu Rajendran	**Dr. H V Ravish Aaradhya**

External Viva

Name of Examiners	Signature with Date
1.	
2.	

DECLARATION

We, **Amrut R(1RV24EC025), Anurag(1RV24EC031), Bhavesh S(1RV24EC044), Chandana B C(1RV24EC053),** students of fourth semester B.E., Department of Electronics and Communication Engineering, RV College of Engineering, Bengaluru, hereby declare that the Design Thinking Lab project titled **'Smart Sort: AI Powered Rapid Grading System for Second-Life Li-ion Cells'** has been carried out by me and submitted in partial fulfillment for the award of degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** during the year 2025-26.

Further, we declare that the content of the report has not been submitted previously by anybody for the award of any degree or diploma to any other university.

I also declare that any Intellectual Property Rights generated out of this Design Thinking Lab carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru
Date: 07-07-2026

Name	Signature
1.Amrut R(1RV24EC025)	
2.Anurag(1RV24EC031)	
3.Bhavesh S(1RV24EC044)	
4.Chandana B C(1RV24EC053)	

ACKNOWLEDGEMENT

We are indebted to our group mentor, **Ms. R. Sindhu Rajendran, Assistant Professor, Department of Electronics and Communication Engineering**, for her wholehearted support, valuable suggestions, and invaluable guidance throughout this Design Thinking Lab project. Her continuous encouragement and advice greatly contributed to the successful completion of this work and the preparation of this report.

We express our sincere gratitude to our group examiner, **Dr. K. B. Ramesh, Associate Professor, Department of Electronics and Communication Engineering**, for his valuable comments and constructive suggestions.

We also express our sincere acknowledgement to our DTL coordinators, **Dr. Chethana G** and **Dr. P. N. Jayanthi, Assistant Professors, Department of Electronics and Communication Engineering**, for their timely instructions and support in coordinating the Design Thinking Lab.

Our sincere thanks go to **Dr. H. V. Ravish Aaradhya, Professor and Head, Department of Electronics and Communication Engineering, RV College of Engineering**, for his support and encouragement.

We express our heartfelt gratitude to our respected Principal, **Dr. K. N. Subramanya**, for his encouragement and appreciation of the Design Thinking Lab.

We thank all the teaching and technical staff of the **Department of Electronics and Communication Engineering, RV College of Engineering**, for their valuable support and assistance.

Finally, we take this opportunity to thank our family members and friends for their constant encouragement and unwavering support throughout this project.

ABSTRACT

The increasing use of electric vehicles, laptops, power tools, and other portable electronic devices has resulted in a growing number of discarded lithium-ion batteries. Although many of these cells still retain sufficient capacity for secondary applications, there is currently no simple, affordable, and rapid method to identify batteries suitable for reuse. Conventional battery evaluation techniques, such as charge-discharge capacity testing and Electrochemical Impedance Spectroscopy (EIS), require either expensive laboratory equipment or several hours of testing, making large-scale battery screening impractical. To address this challenge, **Smart Sort** has been developed as a low-cost, AI-driven battery grading system capable of estimating the true **State of Health (SoH)** of lithium-ion cells within a few seconds.

The proposed system performs **dynamic load testing** by applying a short constant-current load pulse to the battery using an **Arduino Uno**-based test platform comprising an **ADS1115 16-bit Analog-to-Digital Converter (ADC)**, an **LM358 operational amplifier**, an **IRLZ44N MOSFET**, and a **1 Ω shunt resistor**. During the load and recovery phases, the system records the battery's voltage response with high precision. From this response, four physics-based diagnostic features—**Internal Resistance, Maximum Voltage Drop, Recovery Amount, and Initial Recovery Slope**—are extracted. These features provide meaningful insight into the battery's electrochemical behaviour and degradation, enabling more reliable health assessment than conventional open-circuit voltage measurements.

The extracted features are standardized using **StandardScaler** and processed by a **Random Forest classifier** implemented in **Python** using the **Scikit-learn** library. The model was trained using a dataset of **300 Monte Carlo LTspice simulations** representing **Grade A, Grade B, and Grade C** battery conditions. For every battery tested, the system predicts the corresponding health grade along with a confidence score, providing users with an indication of the reliability of the classification. The developed model achieved **100% classification accuracy on the simulated LTspice test dataset**. However, validation using real aged lithium-ion batteries is currently in progress and forms the next stage of the project.

LIST OF TABLES

OVERVIEW

Lithium-ion batteries have become indispensable in modern technology, powering electric vehicles, portable electronics, and renewable energy storage systems. As these batteries reach the end of their primary service life, a significant number of cells still retain sufficient capacity for reuse in less demanding applications. However, identifying such reusable cells remains a major challenge.

Conventional battery evaluation techniques, such as charge-discharge capacity testing and Electrochemical Impedance Spectroscopy (EIS), are either time-consuming or require expensive laboratory equipment. These limitations make large-scale battery screening impractical, resulting in many reusable batteries being unnecessarily discarded and contributing to increasing electronic waste.

To address this challenge, this project proposes **Smart Sort**, an AI-driven rapid battery grading system for second-life lithium-ion cells. Instead of relying solely on static voltage measurements, the system evaluates the battery's dynamic electrical response during a short load-and-recovery test to estimate its State of Health (SoH). This approach enables faster and more reliable battery grading while maintaining a low-cost and portable design.

The proposed system is intended to assist battery recyclers, repair centres, researchers, and renewable energy users in identifying reusable batteries efficiently. By reducing testing time and improving battery screening, Smart Sort promotes second-life battery utilization, supports sustainable battery management, and contributes to the circular economy.

EMPATHY

To understand the challenges associated with battery reuse and disposal, a questionnaire was conducted among students, electronics hobbyists, and users of battery-powered devices. The objective was to understand current battery disposal practices, awareness of second-life batteries, and user expectations from a battery health assessment system.

Questionnaire

The questionnaire focused on battery usage habits, disposal methods, willingness to reuse old batteries, and concerns related to safety, reliability, and performance. The responses revealed that many users dispose of batteries without knowing their actual condition. A majority of the participants also expressed interest in a quick, reliable, and affordable method to assess battery health before deciding whether to reuse or discard a battery.

User Analysis

The survey indicated that users were primarily concerned about battery safety, performance, and cost. Many participants stated that they would be willing to reuse batteries if a reliable method were available to verify their health. Cost savings, environmental sustainability, and confidence in battery safety emerged as the major factors influencing their decision.

Stakeholder Identification

The primary stakeholders identified include battery recycling centers, EV service centers, electronics repair shops, students and hobbyists, renewable energy users, and researchers working on battery management systems. These groups require a simple, affordable, and reliable method for assessing battery health without the need for specialized equipment or extensive technical expertise.

Key Insights

The empathy study highlighted the need for a battery grading system that is rapid, low-cost, easy to operate, and reliable. Users preferred a solution capable of quickly classifying batteries into reusable and non-reusable categories without requiring specialized knowledge or expensive laboratory equipment. These findings directly influenced the design and development of the proposed **Smart Sort** battery grading system.

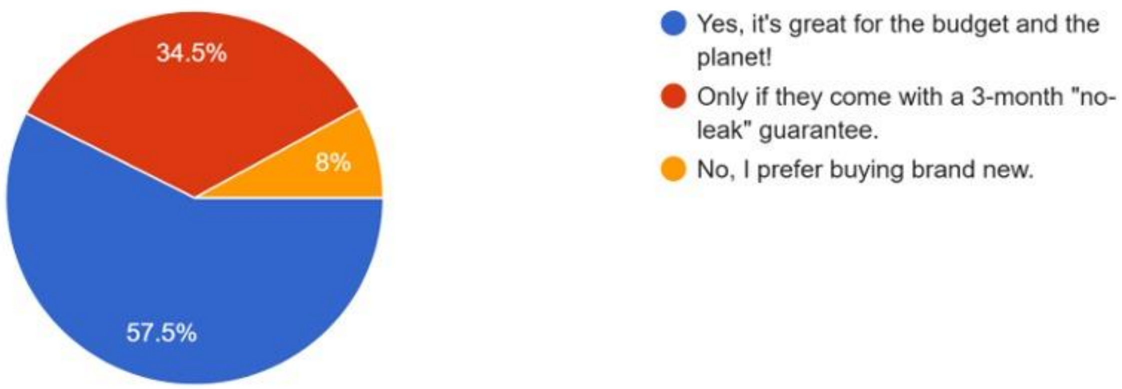

Figure 1:Would you consider using second-life batteries?

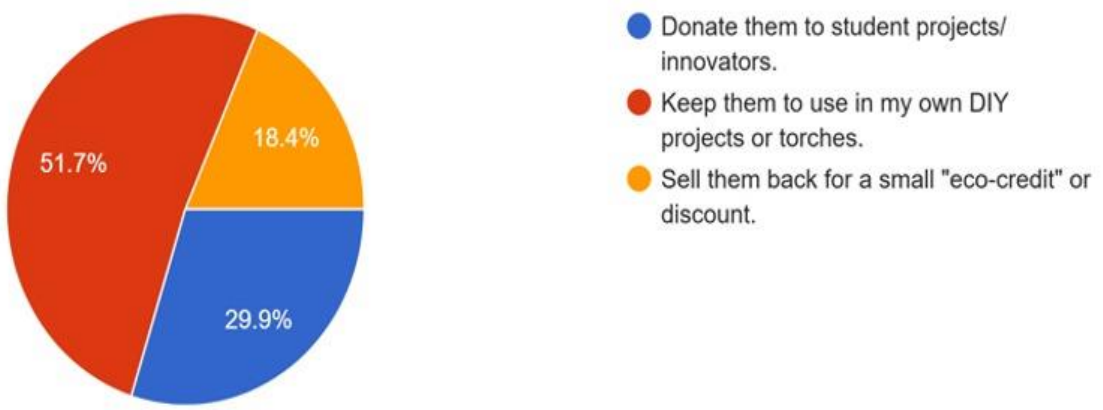

Figure 2:What would you do with old but working batteries?

PROBLEM STATEMENT

A large number of lithium-ion batteries are discarded without assessing their actual health, resulting in the loss of potentially reusable cells and contributing to increased electronic waste. Existing battery grading methods are often time-consuming, expensive, or unsuitable for large-scale screening. Consequently, there is a need for a rapid, low-cost, and reliable battery health assessment system capable of accurately identifying reusable batteries.

IDEATE

Current Status of Art

With the rapid adoption of lithium-ion batteries in electric vehicles, portable electronics, and renewable energy systems, significant research has focused on battery health assessment techniques. Commonly used methods include charge-discharge capacity testing, internal resistance measurement, and Electrochemical Impedance Spectroscopy (EIS). These techniques provide valuable information about battery performance, remaining capacity, and overall State of Health (SoH).

Existing Challenges

Despite their effectiveness, each of these techniques has inherent limitations. Capacity testing requires complete charge-discharge cycles, often taking several hours for a single battery. Electrochemical Impedance Spectroscopy (EIS) offers highly accurate diagnostics but requires specialized laboratory equipment and significant investment.

Although internal resistance measurements are comparatively fast, they alone cannot fully characterize battery degradation. These limitations make large-scale battery screening impractical and frequently result in reusable batteries being discarded unnecessarily.

Proposed Work

To overcome these limitations, **Smart Sort** adopts a dynamic battery testing approach instead of relying on a single static voltage measurement. The system evaluates both the battery's response to a controlled constant-current load pulse and its recovery behaviour after the load is removed.

A constant-current load circuit, built using an **IRLZ44N MOSFET**, an **LM358 operational amplifier**, and a **1 Ω shunt resistor**, applies a controlled load of approximately **1 A** to the battery. An **ADS1115 16-bit Analog-to-Digital Converter (ADC)** measures the battery voltage with high precision, while an **Arduino Uno** controls the testing sequence and transfers the acquired data to a computer for further analysis.

During testing, the battery initially remains at open-circuit conditions. A controlled current pulse is then applied, causing an immediate voltage drop due to the battery's internal resistance. After the load is removed, the battery voltage gradually recovers toward its equilibrium value. This complete load-and-recovery response contains significantly more information about battery health than a simple open-circuit voltage measurement.

The underlying electrochemical behaviour explains this response. The instantaneous voltage drop is primarily governed by the battery's internal resistance and charge-transfer resistance, while the subsequent recovery reflects the relaxation of polarization effects and the diffusion of lithium ions within the electrodes. This recovery follows an RC-like relaxation process, consisting of a rapid initial voltage recovery followed by a slower exponential settling. As lithium-ion batteries age, internal resistance increases, polarization becomes more pronounced, and ion diffusion slows, resulting in larger voltage drops and slower recovery. By analysing this dynamic response, Smart Sort provides a more accurate estimate of the battery's true **State of Health (SoH)** than conventional static voltage measurements.

Feature Extraction

From each load-and-recovery cycle, four physics-based diagnostic features are extracted:

1. **Internal Resistance** – Calculated using Ohm's Law (**R = ΔV / I**) from the instantaneous voltage drop produced by the applied load current.
2. **Maximum Voltage Drop** – The maximum reduction in battery voltage measured between the initial open-circuit voltage and the minimum voltage observed during the load pulse.
3. **Recovery Amount** – The total voltage recovered after the load is removed, representing the battery's ability to return toward its equilibrium state.
4. **Initial Recovery Slope** – The rate of voltage recovery immediately after load removal. This feature reflects battery polarization and ion diffusion characteristics and was found to be the most discriminative feature for distinguishing between Grade A, Grade B, and Grade C batteries during model development.

AI Pipeline

The The extracted features are standardized using **StandardScaler** and provided as input to a **Random Forest** classifier implemented in **Python** using the **Scikit-learn** library. A Random Forest model was selected because battery degradation exhibits highly nonlinear behaviour, while the available feature set is relatively small and structured. Tree-based ensemble methods are well suited to such problems, offering high classification accuracy, robustness to feature interactions, and a lower risk of overfitting compared to deeper neural network models trained on limited datasets.

The classifier predicts the battery health grade (**Grade A, Grade B, or Grade C**) and also provides a confidence score for each prediction, enabling borderline cases to be identified for further evaluation. The model was trained using a dataset of **300 Monte Carlo LTspice simulations**, equally distributed across the three battery grades to represent realistic variations in internal resistance and recovery characteristics associated with different States of Health (SoH).

The developed Random Forest model achieved **100% classification accuracy on the simulated LTspice test dataset**. Since the training and testing data were generated through simulation, this result demonstrates the effectiveness of the proposed feature extraction method and the complete machine learning pipeline. Validation using real, physically aged lithium-ion batteries are currently underway and represents the next stage of the project before evaluating real-world performance.

Objectives

The primary objective of this project is to estimate the true **State of Health (SoH)** of lithium-ion batteries rather than relying solely on open-circuit voltage measurements, which may appear normal even in batteries that have experienced significant capacity fade or increased internal resistance. By analysing the battery's dynamic response during controlled load and recovery phases, **Smart Sort** provides a more accurate assessment of battery health and classifies cells according to their suitability for second-life applications.

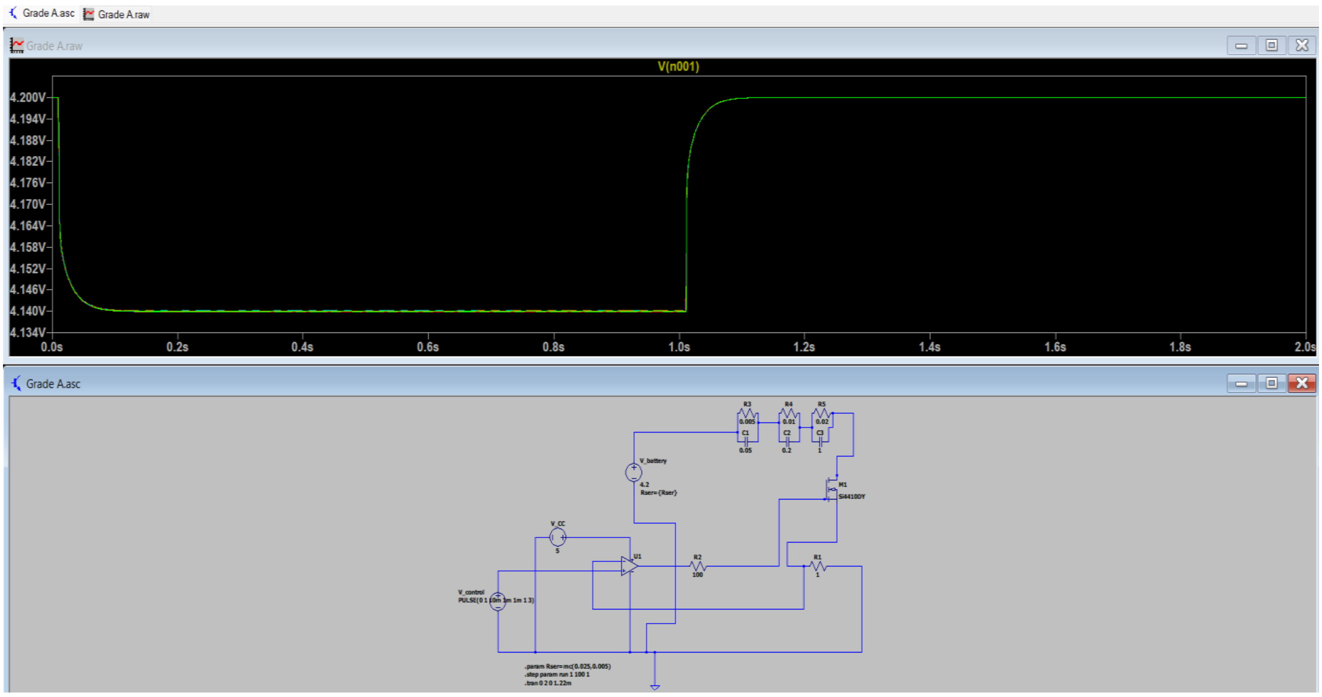

Figure 3: Grade A — shallow voltage sag and fast, near-complete recovery, indicating low internal resistance and healthy diffusion dynamics.

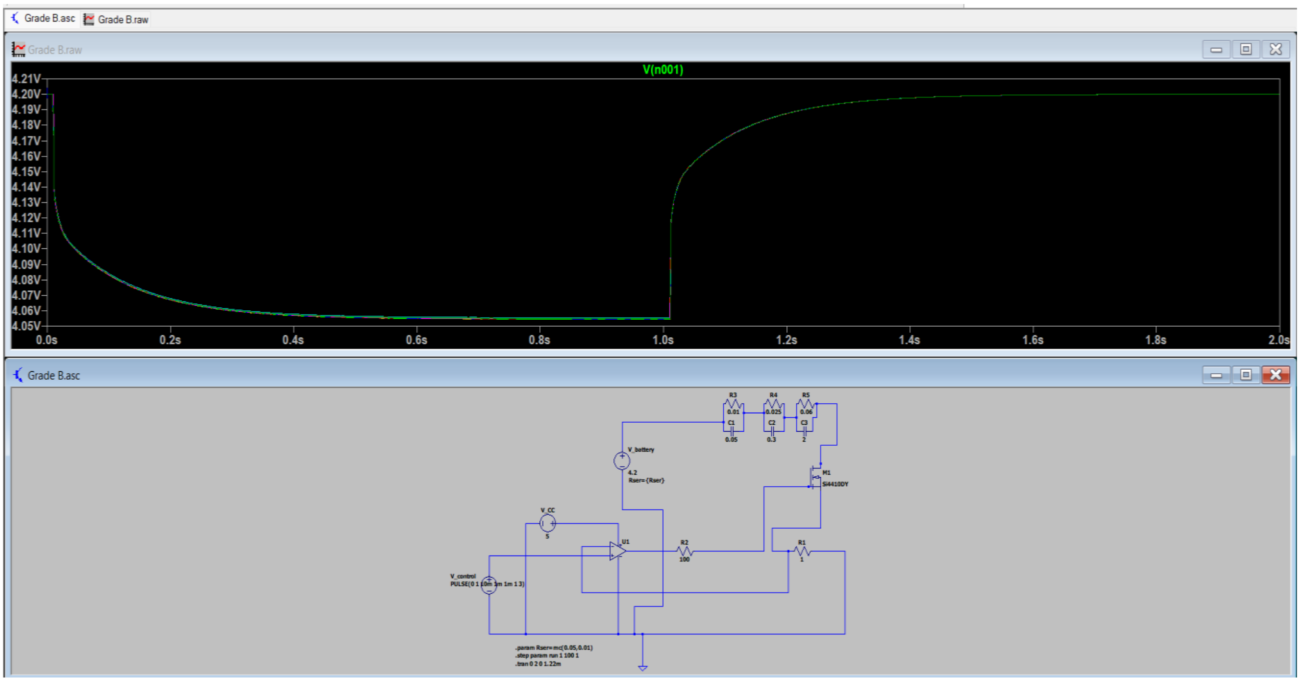

Figure 4: Grade B — a moderate sag and a slower recovery, consistent with a cell that has aged but still retains usable capacity.

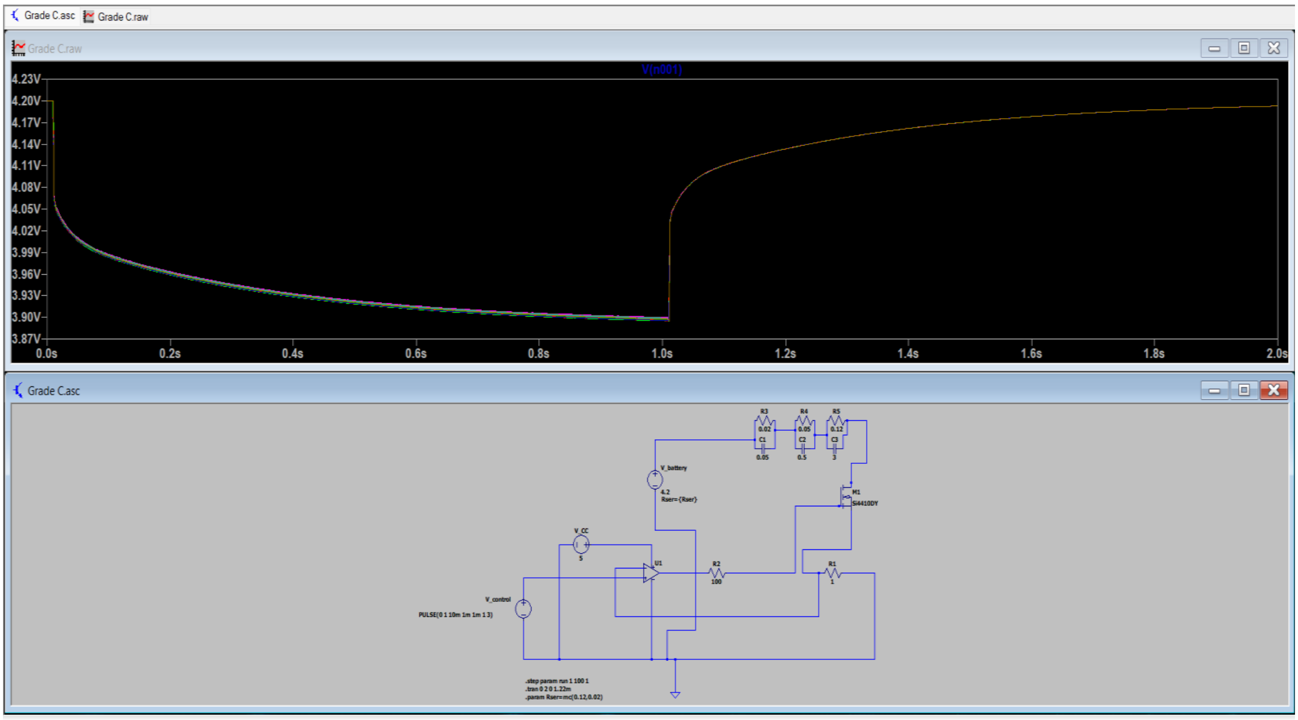

Figure 4: Grade C — a deep sag and sluggish, incomplete recovery, pointing to high internal resistance and degraded diffusion pathways typical of end-of-life cells.

```
PS C:\Users\bhhav\OneDrive\Documents\DTL> python extract_datasets.py
Processing C:\Users\bhhav\OneDrive\Documents\DTL\LTspice\Grade A.txt...
Processing C:\Users\bhhav\OneDrive\Documents\DTL\LTspice\Grade B.txt...
Processing C:\Users\bhhav\OneDrive\Documents\DTL\LTspice\Grade C.txt...
Saved Basic Dataset to C:\Users\bhhav\OneDrive\Documents\DTL\dataset_basic.csv (300
samples)
Saved Enhanced Dataset to C:\Users\bhhav\OneDrive\Documents\DTL\dataset_enhanced.csv
(300 samples)
```

Figure 6: Python Environment Setup, Dataset Loading, and Data Preprocessing Output

```
PS C:\Users\bhhav\OneDrive\Documents\DTL> python train_model_enhanced.py
=== Enhanced Model Evaluation ===
Accuracy : 1.0000
Precision: 1.0000
Recall   : 1.0000
F1-Score : 1.0000

Confusion Matrix:
[[20  0  0]
 [ 0 20  0]
 [ 0  0 20]]

Classification Report:
              precision    recall  f1-score   support

           A       1.00      1.00      1.00        20
           B       1.00      1.00      1.00        20
           C       1.00      1.00      1.00        20

    accuracy                           1.00        60
   macro avg       1.00      1.00      1.00        60
weighted avg       1.00      1.00      1.00        60

Feature Importance:
 - Internal_Resistance: 0.3640
 - Maximum_Voltage_Drop: 0.2306
 - Recovery_Amount: 0.1859
 - Recovery_Slope: 0.2196
================================
```

Figure 5: Random Forest Model Training and Classification Report

```
Feature: Internal Resistance
--------------------------------------------------------------------------------
       count    mean     std      min      25%      50%      75%      max
Grade
A      100.0  0.0600  0.0001  0.0599  0.0600  0.0600  0.0601  0.0602
B      100.0  0.1454  0.0031  0.1445  0.1448  0.1451  0.1453  0.1759
C      100.0  0.3117  0.1838 -1.0109  0.3007  0.3020  0.3031  1.1018

Feature: Maximum Voltage Drop
--------------------------------------------------------------------------------
       count    mean     std      min      25%      50%      75%      max
Grade
A      100.0  0.0600  0.0001  0.0599  0.0600  0.0600  0.0601  0.0602
B      100.0  0.1611  0.1607  0.1445  0.1448  0.1451  0.1453  1.7519
C      100.0  0.4493  1.0684 -2.1104  0.3014  0.3028  0.3039  7.5830

Feature: Recovery Amount
--------------------------------------------------------------------------------
       count    mean     std      min      25%      50%      75%      max
Grade
A      100.0  0.0600  0.0001  0.0599  0.0600  0.0600  0.0601  0.0602
B      100.0  0.1610  0.1597  0.1445  0.1448  0.1451  0.1453  1.7423
C      100.0  0.4395  1.0648 -2.1012  0.2942  0.2955  0.2966  7.5722

Feature: Recovery Slope
--------------------------------------------------------------------------------
       count    mean     std       min      25%      50%      75%       max
Grade
A      100.0  0.3954  0.0027   0.3902  0.3937  0.3947  0.3966   0.4014
B      100.0  0.6198  2.2233 -21.3912  0.8375  0.8412  0.8456   0.8559
C      100.0  2.3972  6.9805 -15.4438  1.4887  1.5003  1.5119  49.0430
```

Figure 8: Statistical Summary of Extracted Features

Fig. 3: Grade A Battery Response

The Grade A battery exhibits a small voltage drop under load and a rapid recovery after load removal. This indicates low internal resistance, efficient electrochemical behavior, and a high State of Health (SoH), making the cell suitable for second-life applications.

Fig. 4: Grade B Battery Response

The Grade B battery shows a moderate voltage drop and a slower recovery compared to Grade A. These characteristics indicate moderate ageing and increased internal resistance, while the battery still retains sufficient capacity for moderate-power applications.

Fig. 5: Grade C Battery Response

The Grade C battery displays a significant voltage drop and slow, incomplete recovery following load removal. These characteristics indicate high internal resistance, poor diffusion behavior, and severe degradation, making the cell unsuitable for reuse.

Fig. 6: Python Environment Setup, Dataset Loading, and Preprocessing

The Python environment was successfully configured, and a dataset comprising **300 Monte Carlo LTspice battery simulations** was generated and processed. The extracted physics-based features were standardized using **StandardScaler**, preparing the data for machine learning model training.

Fig. 7: Random Forest Model Training and Classification Report

The Random Forest classifier achieved **100% classification accuracy** on the simulated LTspice test dataset, with perfect precision, recall, and F1-score across all battery grades. The feature importance analysis confirms that the extracted physics-based features effectively distinguish battery health conditions.

Fig. 8: Statistical Summary

The statistical summary presents the distribution of the four extracted features—Internal Resistance, Maximum Voltage Drop, Recovery Amount, and Recovery Slope—for Grade A, Grade B, and Grade C batteries. The clear separation between the feature values demonstrates the effectiveness of the proposed feature extraction method for battery health classification.

PROTOTYPE

The **Smart Sort** prototype was developed to validate whether a dynamic load-and-recovery test could be implemented using low-cost hardware and used for accurate battery State of Health (SoH) estimation through machine learning. The system consists of two main stages: a hardware stage that performs the battery test and acquires measurement data, and a software stage running on a connected PC that extracts features and classifies the battery.

On the hardware side, the battery under test is connected to a constant-current load circuit built using an **IRLZ44N MOSFET**, an **LM358 operational amplifier**, and a **1**

Ω shunt resistor. The LM358 regulates the MOSFET to maintain a constant load current of approximately **1 A** during testing. An **ADS1115 16-bit Analog-to-Digital**

Converter (ADC) measures both the battery voltage and the voltage across the shunt resistor with high precision, while an **Arduino Uno** controls the timing of the load pulse and transmits the acquired data to the PC via serial communication over the I²C interface.

On the software side, the recorded voltage measurements are processed in **Python**, where four physics-based features—**Internal Resistance, Maximum Voltage Drop, Recovery Amount, and Initial Recovery Slope**—are extracted for each test cycle. These features are standardized using **StandardScaler** and supplied to a trained **Random Forest** classifier, which predicts the battery grade (**Grade A, Grade B, or Grade C**) together with a confidence score. The complete testing process, from applying the load pulse to obtaining the final prediction, is completed within a few seconds, significantly reducing the time required compared to conventional capacity testing methods.

The current prototype is a **bench-top battery measurement and classification system** rather than a fully automated sorting platform. Batteries are inserted and removed manually after testing, and no automatic sorting mechanism has been implemented in the present prototype. Future work includes integrating an automated battery handling and sorting mechanism to physically separate cells based on their predicted health grade.

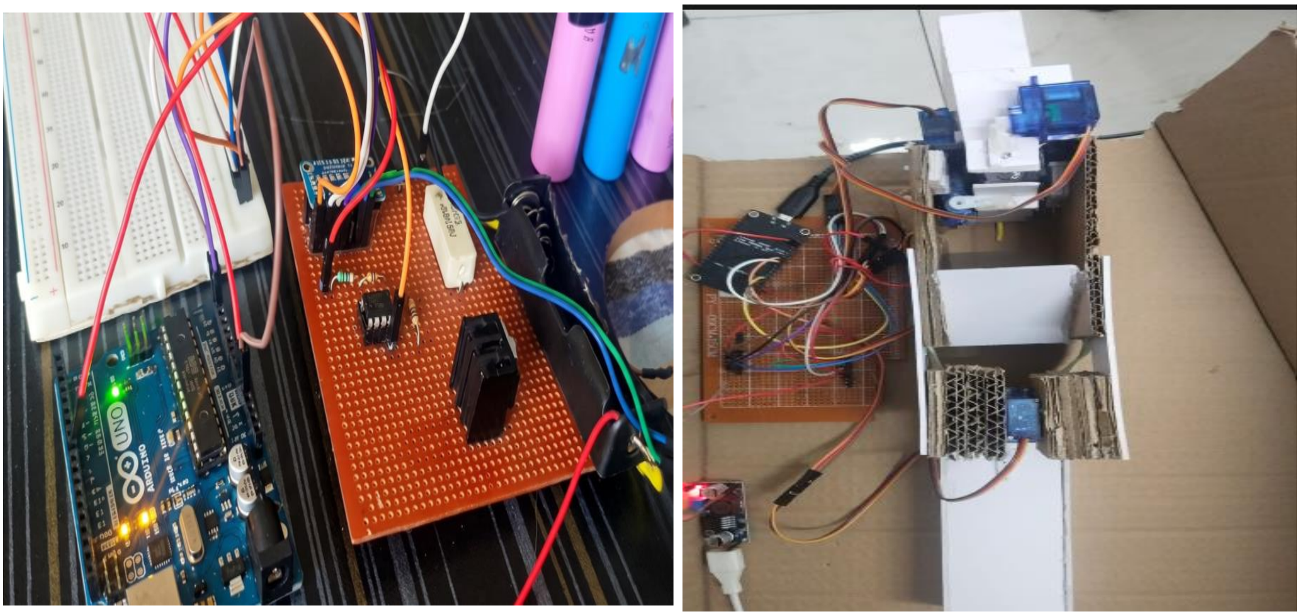

Figure 9:Prototype Hardware Implementation

CONCLUSION

The **Smart Sort** project demonstrates a practical and cost-effective approach for rapid **State of Health (SoH)** estimation of second-life lithium-ion batteries. By combining a constant-current load-and-recovery test with physics-based feature extraction and a **Random Forest** machine learning classifier, the proposed system provides a more reliable assessment of battery health than conventional open-circuit voltage measurements. The prototype, built using an **Arduino Uno**, **ADS1115 16-bit ADC**, **LM358 operational amplifier**, and **IRLZ44N MOSFET**, successfully captures the battery's dynamic response and classifies its health within a few seconds using a Python-based software pipeline.

The developed Random Forest classifier achieved **100% classification accuracy on the simulated LTspice test dataset**, demonstrating the effectiveness of the proposed feature extraction method and the overall machine learning pipeline. Validation using real, physically aged lithium-ion batteries remains the next stage of the project to establish real-world performance. Overall, the proposed system provides a promising low-cost solution for rapid battery grading, supporting the reuse of second-life batteries, reducing electronic waste, and promoting sustainable battery management.

FUTURE SCOPE

The immediate priority is to validate the proposed system using real, physically aged lithium-ion batteries to evaluate its performance beyond simulation-based datasets. Expanding the dataset to include batteries of different chemistries, manufacturers, capacities, and ageing conditions will further improve the robustness and generalization capability of the machine learning model.

Future hardware enhancements include the development of an automated battery sorting mechanism capable of physically separating cells according to their predicted health grade, thereby eliminating manual handling and increasing testing throughput. Additional improvements include cloud-based storage of battery test records, remote monitoring for battery recycling centres operating multiple testing units, and long-term battery performance tracking to compare predicted battery health with actual field performance. With these enhancements, **Smart Sort** has the potential to evolve into a practical battery grading system for recycling facilities, repair centres, research laboratories, and second-life energy storage applications.

REFERENCES

[1] Natella, D., Onori, S., and Vasca, F., "A Co-Estimation Framework for State of Charge and Parameters of Lithium-Ion Battery with Robustness to Aging and Usage Conditions," IEEE Transactions on Industrial Electronics, vol. 69, no. 12, pp. 12489–12499, 2022.

[2] Lai, X., Yuan, M., Tang, X., Yao, Y., Weng, J., Gao, F., Ma, W., and Zheng, Y., "Co-Estimation of State of Charge and State of Health for Lithium-Ion Batteries Using an Enhanced Electrochemical Model," IEEE Transactions on Power Electronics, vol. 36, no. 6, pp. 6302–6313, 2021.

[3] Heeger, D., Partridge, M., Trullinger, V., and Wesolowski, D., Lithium Battery Health and Capacity: Estimation Techniques Using Embedded Electronics, Sandia National Laboratories, SAND2017-10722, 2017.

[4] De Angelis, A., Buchicchio, E., Santoni, F., and Carbone, P., "Practical Broadband Measurement of Battery Electrochemical Impedance Spectroscopy," IEEE Transactions on Instrumentation and Measurement, vol. 70, pp. 1–10, 2021.

[5] Sun, Y., Smith, K., and Mohan, A., "Evaluating the Use of Rapid DC Pulses for Diagnosing Lithium-Ion Batteries," National Renewable Energy Laboratory (NREL), 2025.

[6] Gao, Z., Lan, T., Yin, H., and Liu, Y., "A Robust Machine Learning-Based System for Battery Grading and Lifecycle Prediction for Electric Vehicle Applications," International Journal of Information Technology, Springer Nature, 2025.

[7] Kim, J., Park, S., Lee, H., and Choi, W., "AI-Integrated Smart Grading System for End-of-Life Lithium-Ion Batteries Based on Multi-Parameter Diagnostics," Energies, MDPI, vol. 18, no. 22, pp. 5915, 2025.

[8] Rout, S., Samal, S. K., and Gelmecha, D. J., "Estimation of State of Health for Lithium-Ion Batteries Using Advanced Data-Driven Techniques," Scientific Reports, Nature, vol. 15, pp. 30438, 2025.

[9] Kumar, R. S., and Narayana, P. L., "Hybrid Machine Learning Framework for Predictive Maintenance and Anomaly Detection in Lithium-Ion Batteries Using Enhanced Random Forest," Scientific Reports, Nature, 2025.

[10] Wang, P., Liu, G., and Bage, A. N., "Multi-Modal Framework for Battery State of Health Evaluation Using Open-Source Electric Vehicle Data," Nature Communications, vol. 16, pp. 1137, 2025.

[11] Santos, A. L., Alves, W., and Ferreira, P., "Challenges Faced by Lithium-Ion Batteries in Effective Waste Management," Sustainability, vol. 17, no. 7, pp. 2893, 2025.